Lab 5 Part 2

~two interactions:

① @ night ⟶ to give it a job

② in morning ⟶ to get result.

~which approach:

① sense all night: Pi watches, listens, or measures overnight + keeps a log

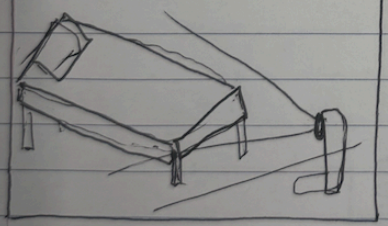
② process all night: give it existing data (a day's photos, a long recording) + it works through it overnight

!! RULES !!

① take a sample every 5 to 30 secs instead of streaming continuously

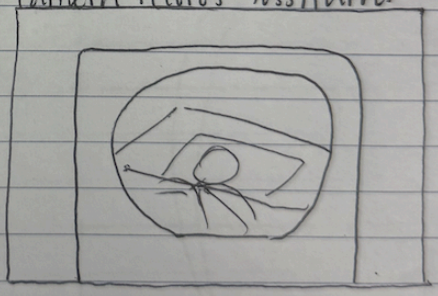
② keep all data on pi: tell roommate, point camera away from bed, delete snapshots afterwards.

position camera near bed

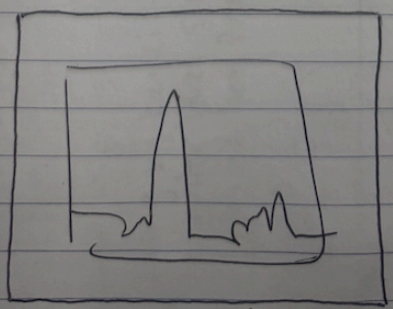


Before bed

camera records toss/turn.

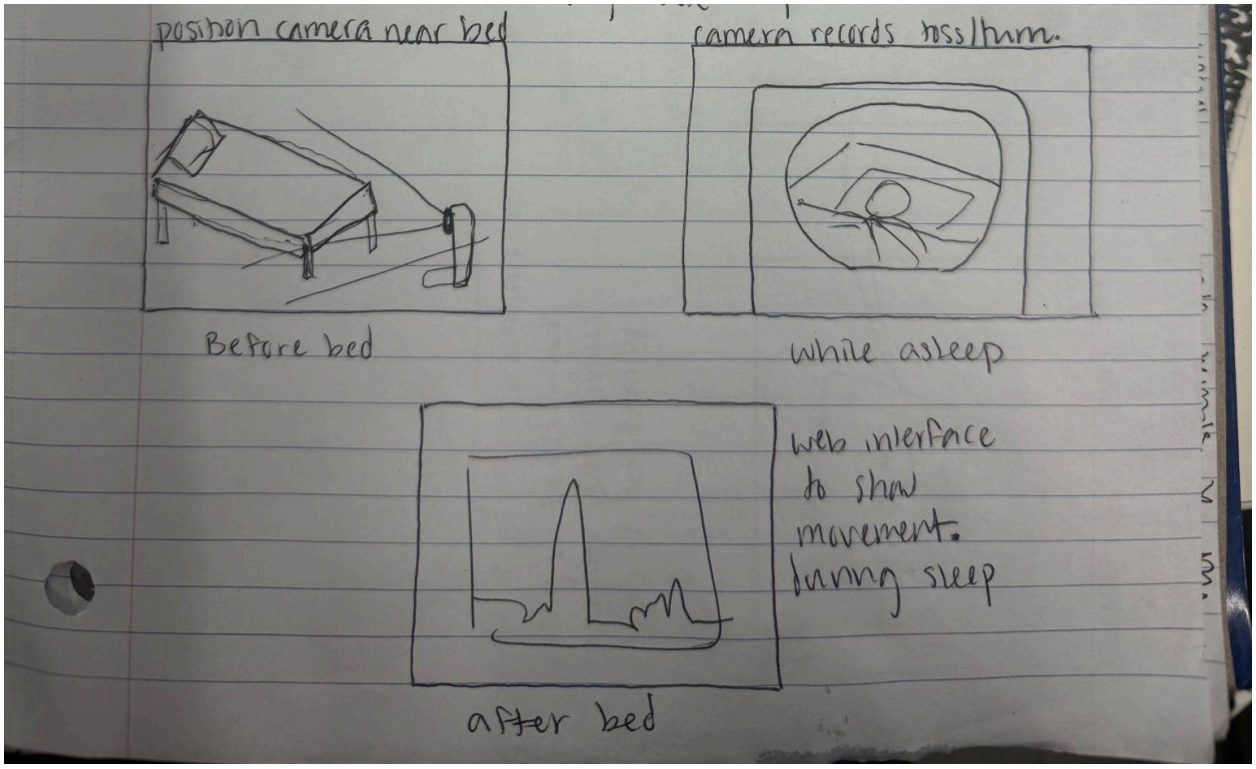


while asleep



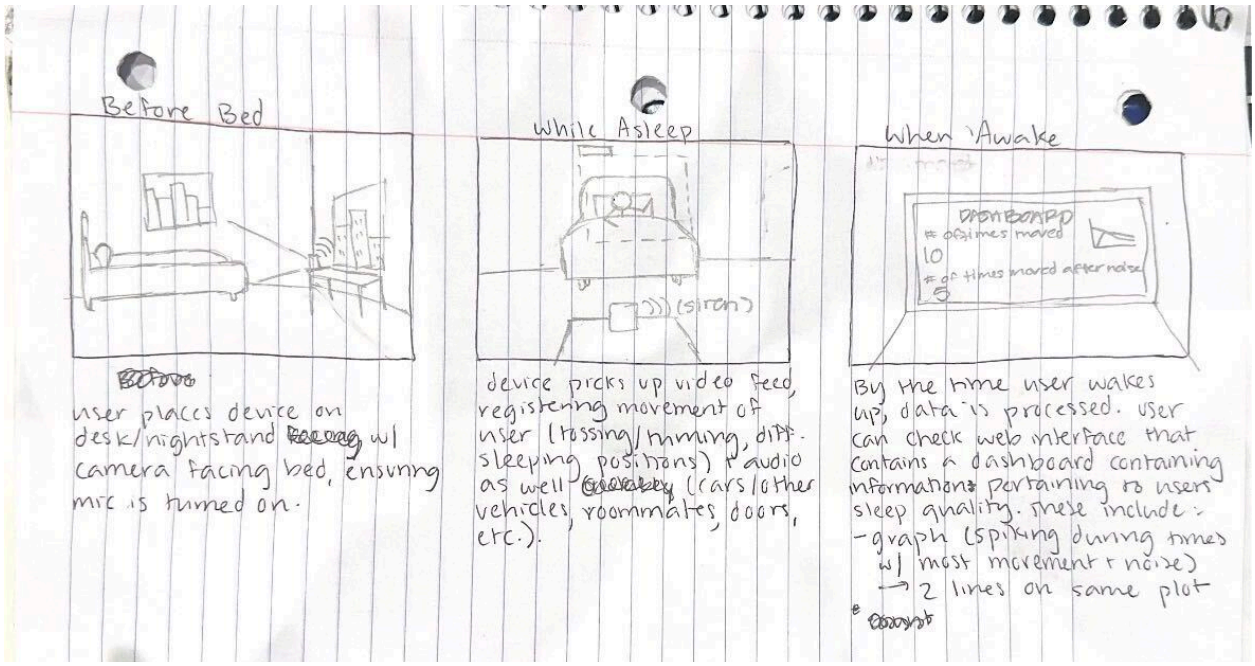
after bed

web interface to show movement during sleep



Our initial storyboards were not as creative as we could have made them. When coming up with our ideas we were thinking very small and 1-dimensional: we were so focused on capturing movement the user was making during their sleep and reporting it to them. We didn't recognize how we could make a story that is 1) worthy of being captured and processed and 2) would be worth it to be purchased. The important question we needed to ask was "Why us?". Why should a user go out of their way to purchase our device?

Revised Storyboard:



Our revised storyboard aims to explore a combination of topics that we individually affect quality of sleep and also provide useful information about a person's surroundings to enable them to make meaningful decisions to improve them.

We already knew [from our storyboards] that we wanted to capture live data by seeing the frequency and times a person tosses and turns throughout the night. However, we wanted to take this further and see what qualities would make a person want to purchase a separate device, as the number of times a person tosses and turns during a night's sleep isn't indicative of anything and doesn't provide any useful information in and of itself and, additionally, doesn't drive the user in any direction to enable them to make an informed decision on what they can change about their sleeping habits and personal life.

Someone mentioned that at the House their room faces the Queensboro Bridge, which can be pretty distracting at night with all of the lights on the bridge, not to mention the lights and sounds from the cars themselves. We then realized that there could be a correlation between how much light a person is exposed to when trying to rest and the amount of tossing and turning being done could be correlated. We decided for our visualization to be a graph that shows the amount of tossing and turning in one color on the graph and the amount of light exposure the device got on the other.

After testing the device, we recognized that moments with the most light exposure induced bigger bits of movement during sleep. This information can enable users to make lifestyle changes such as adding curtains, repositioning beds, reorganizing rooms, and even maybe move out (to find a unit they are better suited for).